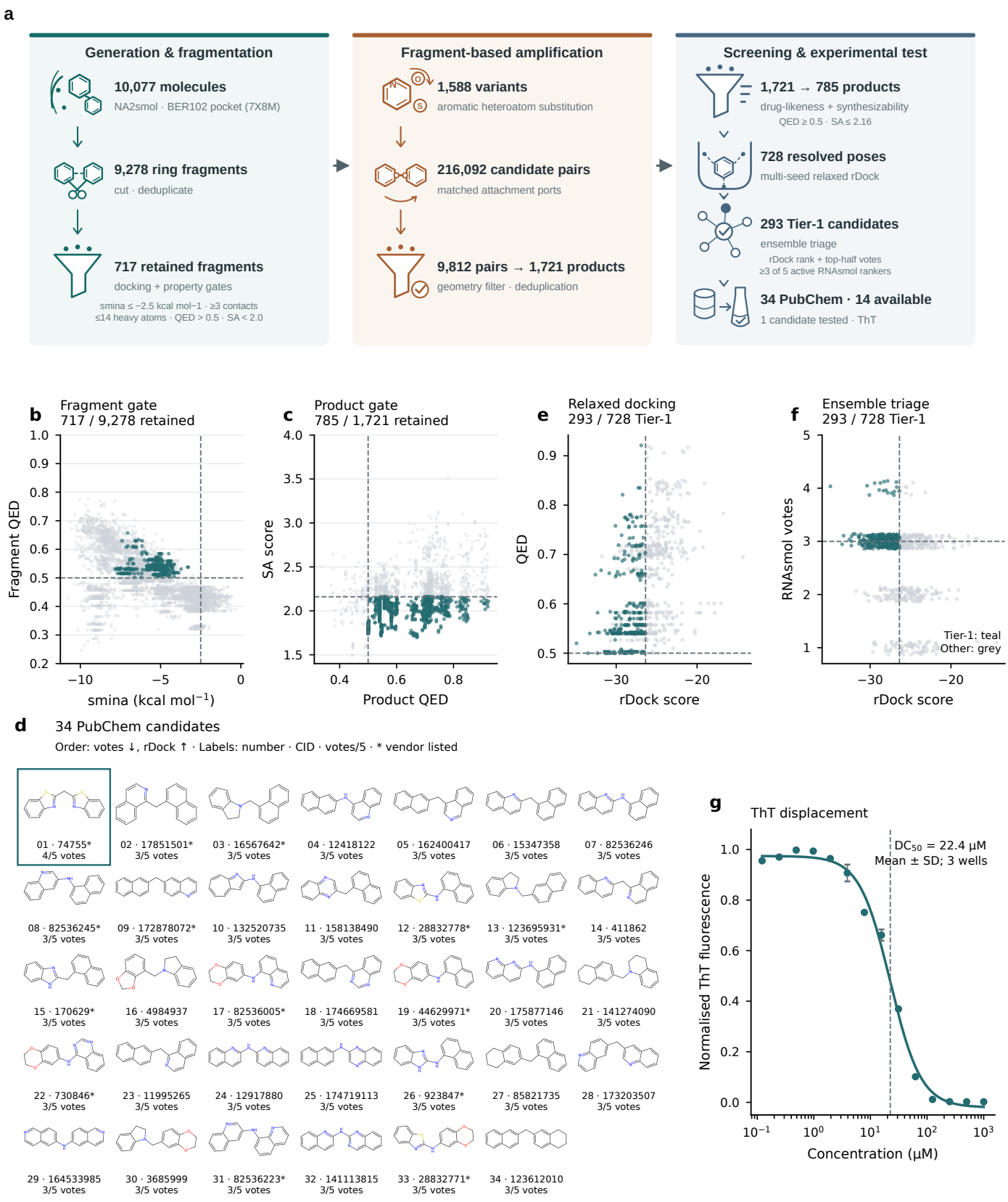

a
Generation & fragmentation
10,077 molecules
NA2smol · BER102 pocket (7X8M)
9,278 ring fragments
cut · deduplicate
717 retained fragments
docking + property gates
smina ≤ −2.5 kcal mol−1 · ≥3 contacts
≤14 heavy atoms · QED > 0.5 · SA < 2.0
Fragment-based amplification
1,588 variants
aromatic heteroatom substitution
216,092 candidate pairs
matched attachment ports
9,812 pairs → 1,721 products
geometry filter · deduplication
Screening & experimental test
1,721 → 785 products
drug-likeness + synthesizability
QED ≥ 0.5 · SA ≤ 2.16
728 resolved poses
multi-seed relaxed rDock
293 Tier-1 candidates
ensemble triage
rDock rank + top-half votes
≥3 of 5 active RNAsmol rankers
34 PubChem · 14 available
1 candidate tested · ThT
b
Fragment gate
717 / 9,278 retained
Fragment QED
smina (kcal mol−1)
c
Product gate
785 / 1,721 retained
SA score
Product QED
e
Relaxed docking
293 / 728 Tier-1
QED
rDock score
f
Ensemble triage
293 / 728 Tier-1
RNAsmol votes
rDock score
Tier-1: teal
Other: grey
d
34 PubChem candidates
Order: votes ↓, rDock ↑ · Labels: number · CID · votes/5 · * vendor listed
01 · 74755* 4/5 votes
02 · 17851501* 3/5 votes
03 · 16567642* 3/5 votes
04 · 12418122 3/5 votes
05 · 162400417 3/5 votes
06 · 15347358 3/5 votes
07 · 82536246 3/5 votes
08 · 82536245* 3/5 votes
09 · 172878072* 3/5 votes
10 · 132520735 3/5 votes
11 · 158138490 3/5 votes
12 · 28832778* 3/5 votes
13 · 123695931* 3/5 votes
14 · 411862 3/5 votes
15 · 170629* 3/5 votes
16 · 4984937 3/5 votes
17 · 82536005* 3/5 votes
18 · 174669581 3/5 votes
19 · 44629971* 3/5 votes
20 · 175877146 3/5 votes
21 · 141274090 3/5 votes
22 · 730846* 3/5 votes
23 · 11995265 3/5 votes
24 · 12917880 3/5 votes
25 · 174719113 3/5 votes
26 · 923847* 3/5 votes
27 · 85821735 3/5 votes
28 · 173203507 3/5 votes
29 · 164533985 3/5 votes
30 · 3685999 3/5 votes
31 · 82536223* 3/5 votes
32 · 141113815 3/5 votes
33 · 28832771* 3/5 votes
34 · 123612010 3/5 votes
g
ThT displacement
DC50 = 22.4 µM
Mean ± SD; 3 wells
Normalised ThT fluorescence
Concentration (µM)